

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: pBR322 derived plasmids containing retroviral vectors.
Species Used	Mouse, NIH/3T3 derived retroviral vector packaging cell lines		

Before the Pulse

Cell Growth Medium	DMEM + 10% Fetal Bovine Serum + 1% Glutamine (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Log phase
		Pre-pulse Incubation	5 min, with DNA at room temperature
Wash Solution	Phosphate Buffered Saline		

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	Room temperature		
Electroporation Medium*	DMEM + 10% Fetal Bovine Serum + 1% Glutamine	Cuvette Gap	0.4 cm
Cell Density	1.5 x 10 ⁶ (6) cells / ml	Voltage	0.2 kV
Volume of Cells	0.5 ml	Field Strength	0.5 kV/cm
DNA Concentration	0.1 to 1 µg / µl		
DNA Resuspension Buffer	Not given	Capacitor	960 µF
Volume of DNA	10 µl	Resistor	(Pulse Controller) Ω none
		Time Constant	20 to 25 msec

After the Pulse

Outgrowth Medium	DMEM + 10% Fetal Bovine Serum + 1% Glutamine
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Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	Not given
Selection Method or Assay Used	G418
Electroporation Efficiency	50 to 200 transformants / µg DNA
Per Cent Survival	Not given

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Survey Number

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